Microsoft Elevate

CAPSTONE PROJECT

TRACKING DIGITAL INDIA:

Analytics and Forecasting of AI Adoption Among Indian Students.

PRESENTED BY

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PROBLEM STATEMENT:

Currently, Artificial Intelligence tools are rapidly being introduced into the educational ecosystem, fundamentally changing how Indian students across various streams approach tasks like assignments, doubt solving, and coding help. It is important to accurately track and understand this digital transition to ensure equitable access and to gauge the current awareness and trust levels among the student population. Eventually, addressing the clear disparities in digital infrastructure—specifically the high percentages of students experiencing poor (32.65%) or medium (35.65%) internet connectivity —alongside the friction of academic acceptance from professors, becomes a major concern for educational institutions. The crucial part is the comprehensive analysis and forecasting of these AI adoption trends to effectively bridge the digital divide and support the broader goals of Tracking Digital India.

PROPOSED SOLUTION:

The proposed system aims to address the challenge of analysing and forecasting AI adoption among Indian students. This involves leveraging data analytics and visualization techniques to forecast usage and demand patterns accurately. The solution will consist of the following components:

Data Collection:

- Gather dataset records on 2000 surveyed students, including their preferred AI tools, average daily usage hours, and academic streams.
- Utilize these data sources to evaluate relevant factors like internet access levels (Poor, Medium, High) and device usage (Mobile, Tablet, Laptop) to enhance analytics.

Data Preprocessing:

- Clean and preprocess the collected Kaggle data using Power Query to handle missing values, outliers, and inconsistencies.
- Feature engineering to extract relevant features from the data, such as parsing specific use cases like assignments, coding help, and doubt solving, that might impact AI demand.

Machine Learning Algorithm (Data Modelling & Forecasting):

- Implement data modelling and forecasting features within Power BI to analyse trends based on historical patterns of AI awareness and trust.
- Consider incorporating other factors like professors' allowance of AI and students' willingness to pay to improve prediction accuracy.

Deployment:

- Develop a user-friendly Power BI dashboard interface that provides comprehensive insights into the usage of tools like ChatGPT, Copilot, and Gemini.
- Deploy the solution on a scalable and reliable platform (Power BI Service), considering factors like user accessibility for tracking Digital India metrics.

Evaluation:

- Assess the dashboard's performance using appropriate metrics to effectively visualize key data points, such as the 5.73 average awareness level out of 10.
- Fine-tune the visual model based on feedback and continuous monitoring of the analytical insights.

SYSTEM APPROACH:

The "System Approach" section outlines the overall strategy and methodology for developing and implementing the AI adoption analytics and forecasting dashboard for Indian students.

System requirements:

- **Software:** Microsoft Power BI Desktop for data modelling and visualization, Microsoft Excel or a text editor for initial Kaggle dataset inspection, and a modern web browser for Power BI Service deployment.
- **Hardware:** A standard computer or laptop capable of smoothly running Power BI Desktop (e.g., Windows 10/11 OS, minimum 4GB RAM, 8GB+ recommended for efficiently processing the 2000 surveyed student records).
- **Data Source:** Downloaded Kaggle dataset containing the raw data on student AI preferences, internet access levels, and device usage.

Library required to build the model:

- **Power Query (M Language):** The essential data preparation technology required to clean, transform, and load the raw Kaggle dataset into Power BI.

- **DAX (Data Analysis Expressions):** The primary formula language required to build custom calculated columns and measures, enabling the calculation of specific metrics like the 2.99 average trust in AI and the 5.73 average awareness level.

- **Power BI Standard Visuals Library:** The built-in visualization components necessary to render the dashboard charts, such as mapping the preferred AI tools (ChatGPT, Copilot, Gemini, etc.) and analysing the distribution across academic streams (Science, Arts, Commerce, etc.).

ALGORITHM & DEPLOYMENT:

In the Algorithm section, we describe the analytical and forecasting models chosen for analysing AI adoption trends among students. Here is the structure for this section:

Algorithm Selection:

- Provide a brief overview of the chosen algorithm: For this dashboard, descriptive analytics and Power BI's built-in statistical/forecasting models (such as exponential smoothing for trend analysis) were selected. This is justified as the problem statement requires interpreting survey data to understand current adoption patterns and forecasting future growth across different academic streams.

Data Input:

- Specify the input features used by the algorithm: The model ingests data from 2000 surveyed students , tracking their preferred AI tools such as ChatGPT (433 users) , Copilot (432 users) , and Gemini (424 users) .
- Additional relevant factors include device usage (Mobile, Tablet, Laptop) , specific use cases like Assignments and Coding Help , and internet access levels .

Training Process:

- Explain how the algorithm is trained: In a Power BI environment, the "training" phase translates to data modelling—establishing relationships within the Kaggle dataset to link student demographics to their usage habits.
- Specific techniques employed include writing customized DAX (Data Analysis Expressions) measures to accurately calculate and aggregate critical KPIs, such as the 5.73 average awareness level and the 2.99 average trust in AI.

Prediction Process:

- Detail how the trained algorithm makes predictions: The analytical model forecasts future AI adoption curves by extrapolating current sentiment data, heavily weighting the metric that 51.3% of students are already willing to pay for AI tools.
- Real-world variables considered during the prediction phase include institutional friction, specifically accounting for the fact that 50.95% of professors currently do not allow AI, which will shape the trajectory of future academic adoption.

RESULT:

Tracking Digital India
Analytics and Forecasting of AI Adoption Among Indian Students
2000
Total Students Surveyed
5.73
Avg. Awareness Lv. Out of 10
2.99
Avg. Trust in AI Out of 5
2.64
Avg. Daily Usage Hrs.
Willing to Pay for AI Tools
No 48.7%
Yes 51.3%
Preferred AI Tools
ChatGPT 433
Copilot 432
Gemini 424
Other 403
Claude 157
Bard 151
Use Cases
Select all
Assignments
Assignments, Coding Help
Assignments, Coding Help, Content Writing
Assignments, Coding Help, Doubt Solving
Assignments, Coding Help, Exam Preparation
Assignments, Coding Help, MCQ Practice
Do Prof. Allow AI
Yes 49.05%
No 50.95%
Device Used
Mobile 32.2%
Laptop 34.3%
Tablet 33.5%
Stream
Law 154
Science 260
Pharmacy 158
Hotel-management 165
Arts 253
Agriculture 172
Commerce 244
Medical 176
Management 211
Engineering 207
Internet Access Level
Medium 35.65%
High 31.7%
Poor 32.65%

CONCLUSION:

The Power BI dashboard effectively analyzes and forecasts AI adoption among 2000 surveyed Indian students, providing clear insights into current digital trends. Key findings reveal a moderate average AI awareness level of 5.73 out of 10, alongside an average trust score of 2.99. The visual data clearly establishes that ChatGPT, Copilot, and Gemini are the most preferred AI tools, currently utilized by 433, 432, and 424 students respectively.

A significant challenge highlighted during this analysis is the stark infrastructural divide and institutional friction currently present. Specifically, 32.65% of surveyed students report poor internet access, while an additional 35.65% have only medium access. Furthermore, despite 51.3% of students expressing a willingness to pay for premium AI tools, there remains a substantial barrier in academic acceptance, with 50.95% of professors currently not allowing AI usage in their coursework. Potential improvements for this analytical solution could include integrating real-time data feeds to continually monitor how these specific adoption barriers evolve across different academic streams over time.

Ultimately, this project emphasizes the crucial importance of accurate analytics and forecasting of AI adoption within the education sector. Understanding these specific demographic patterns, device preferences, and access limitations is essential for stakeholders to ensure a stable, equitable, and effective integration of digital tools across India.

FUTURE SCOPE:

Discussing potential enhancements and expansions for the system is vital for the next phase of this analytics project. Future iterations could include incorporating additional data sources to expand beyond the initial 2000 surveyed students, perhaps tracking adoption year-over-year. Expanding the system to cover multiple cities or regions would provide a much deeper understanding of how varying internet access levels—currently showing 32.65% with poor access and 35.65% with medium access—impact adoption across diverse academic streams like Science, Commerce, and Arts. Additionally, tracking this data over time could help in optimizing the algorithm for better performance in forecasting specific metrics, such as the 2.64 average daily usage hours. Furthermore, future development should consider the integration of emerging technologies, such as advanced machine learning techniques, to build more robust predictive models. These models could dynamically forecast shifts in preferred AI tools, predicting how quickly alternatives like Gemini or Claude might close the gap with current leaders like ChatGPT and Copilot. Incorporating concepts like edge computing could also be explored to facilitate localized data collection and analysis for the significant portion of students facing connectivity barriers, ensuring a more comprehensive approach to Tracking Digital India.

REFERENCES:

Dataset: AI Tool Usage by Indian College Students 2025, sourced from Kaggle.com. This dataset was instrumental in developing the analytics for AI adoption among students.

Link: https://www.kaggle.com/datasets/rakeshkapilavai/ai-tool-usage-by-indian-college-students-2025

GitHub Link: https://github.com/Prob2005/PowerBI-AI-Adoption-Among-Indian-Students-2025

Thank You